Circuit Lab, Division B

Practice Test 4 (State level) · written test · 50 minutes · 50 questions

Team name: _____ Team number: _____

Names: _____

Instructions

- Write all answers on the answer sheet at the back. Only the answer sheet is graded.
- Give numeric answers in metric (SI) units with the correct number of significant figures.
- You may use two calculators and your own paper notes.
- Use $k = 8.99 \times 10^{9}\ \text{N·m}^2/\text{C}^2$ and $e = 1.602 \times 10^{-19}\ \text{C}$. Wires and meters are ideal unless the question says otherwise.
- In logic questions, A′ means NOT A, AB means A AND B, and A + B means A OR B.

Section	Questions
Section I: Multiple choice	1–17
Section II: Short answer and calculations	18–38
Section III: Circuit problems	39–50

Section I: Multiple choice

Circle the one best answer. 1 point each.

1. The voltaic cells of the 1820s gave a current that drifted. For his 1826 experiments, Ohm used a much steadier source. What was it?
   A. A static-electricity machine
   B. A hand-cranked generator
   C. A lemon battery
   D. A thermocouple (copper and bismuth, one junction hot, one cold)

2. Volta's pile (1800) was a stack of disc pairs separated by cloth or cardboard soaked in salt water. The discs in each pair were:
   A. Zinc and copper (or silver)
   B. Iron and lead
   C. Gold and platinum
   D. Carbon and mercury

3. In 1937, who showed that relay and switch circuits follow Boole's algebra of logic?
   A. Heinrich Hertz
   B. Claude Shannon
   C. Gustav Kirchhoff
   D. Georg Ohm

4. Since the SI revision of 2019, the ampere is defined by fixing the exact value of:
   A. The mass of a platinum cylinder
   B. The elementary charge e
   C. The resistance of a mercury column
   D. The speed of sound in air

5. Teflon is near the negative end of the triboelectric series and wool is near the positive end. A Teflon rod is rubbed with wool. Afterward:
   A. The Teflon is positive and the wool is negative
   B. Both are negative
   C. The Teflon is negative and the wool is positive
   D. Neither is charged, because both are insulators

6. Two equal positive point charges are fixed 10 cm apart. What is the electric field at the point exactly halfway between them?
   A. Zero
   B. Twice the field of one charge alone
   C. It points toward the left charge
   D. It points straight away from the line joining them

7. Four 3 V coin cells in series give 12 V, but they cannot crank a car engine. What is the main reason?
   A. Coin cells make AC, not DC
   B. Their internal resistance is far too high to give hundreds of amps
   C. 12 V from coin cells is a different kind of voltage
   D. Coin cells only work at low temperature

8. Which current through a person's chest is most likely to cause ventricular fibrillation (the heart stops pumping properly)?

A. 1 mA
B. 10 mA
C. 100 mA
D. 1 µA

9. One kilowatt-hour (1.00 kWh) of energy is equal to:

A. 3.60 MJ
B. 3.60 kJ
C. 1000 J
D. 60.0 kJ

10. A 9.0 V battery is in series with resistors R1 and R2. R2 has broken open inside. An ideal voltmeter is placed across R2. It reads:

A. 9.0 V
B. 0 V
C. 4.5 V
D. It depends on the value of R1

11. Not counting the two coil pins, how many switch (contact) terminals does an SPDT relay have?

A. 2
B. 3
C. 4
D. 6

12. An AFCI (arc-fault circuit interrupter) breaker is designed to trip when:

A. About 5 mA leaks from the hot wire to ground
B. It senses arcing, such as from a damaged cord or a loose connection
C. A lightning surge raises the voltage for a few microseconds
D. The outlet has no ground wire

13. What happens when you press the TEST button on a GFCI outlet?

A. A small current is sent around the sensor through a resistor; the imbalance makes it trip
B. The hot wire is shorted to the ground wire with a 0 ohm wire
C. The breaker in the main panel is switched off
D. The outlet measures the resistance of the ground wire

14. In a simple zener-diode voltage regulator, the zener diode is connected:

A. Forward biased, in series with the load
B. Reverse biased, in parallel with the load
C. Reverse biased, in series with the load
D. Forward biased, in parallel with the load

15. How many select inputs does a 4-to-1 multiplexer need?

A. 1
B. 2
C. 3
D. 4

16. By De Morgan's law, $(A + B')'$ is equal to:

A. $A'B$
B. $A' + B$
C. AB'
D. $A'B'$

17. A student finishes a current measurement, leaves the red lead in the 10 A jack, turns the dial to volts and touches the leads across a battery. The most likely result is:

A. The meter reads the battery voltage correctly

B. The meter is almost a short circuit across the battery; a large current flows and the meter fuse may blow

C. The meter reads 0 V and nothing else happens

D. The meter shows the battery's internal resistance

Section II: Short answer and calculations

Write the answer with units and the correct number of significant figures. 2 points each.

18. A +1.2 μC charge and a −4.5 μC charge are 18 cm apart. Find the size of the force between them. Is it attractive or repulsive?

19. A balloon has a charge of −32 nC. How many extra electrons does it carry?

20. A +2.0 μC charge is at x = 0 and a +8.0 μC charge is at x = 0.60 m. At what point on the x-axis is the electric field zero?

21. A sine-wave AC voltage measures 34 V peak-to-peak. What is its rms voltage?

22. A drone battery has 3 cells in series (3S). Each cell is 3.6 V, 3400 mAh. In flight the drone draws a steady 12 A. Find the pack voltage and the flight time in minutes.

23. An aluminum wire is 45.0 m long with a diameter of 1.50 mm. Aluminum has $\rho = 2.65 \times 10^{-8}$ Ω·m. Find its resistance.

24. A 1.20 kW space heater runs 6.0 hours a day for the 31 days of January. Electricity costs $0.14 per kWh. Find the cost to the nearest dollar.

25. A 24.0 Ω heating element is connected to 12.0 V for 5.00 minutes. How much heat energy does it give off?

26. A 5-band resistor reads green, blue, black, brown, red. Give its value and tolerance.

27. Two surface-mount (SMD) resistors are marked 683 and 2R2. Give the value of each.

28. Find the equivalent resistance of a 47 Ω resistor in series with (100 Ω parallel with 150 Ω), in series with a 33 Ω resistor.

29. A meter movement reads full scale at 2.00 mA and has a resistance of 50.0 Ω. What shunt resistor, placed in parallel with it, turns it into a 0 to 1.00 A ammeter?

30. A current of 90.0 mA enters a parallel pair of 220 Ω and 330 Ω. How much current flows in the 330 Ω resistor?

31. A 1000 W microwave and a 650 W toaster run at the same time on one 120 V circuit. Find the total current. Which is the smallest standard fuse (10, 15, 20 or 30 A) that will not blow?

32. Three white LEDs (V_F = 3.2 V each) are in series with one resistor on a 12.0 V supply. They must run at 20. mA. Find the resistor and the power it uses.

33. Find the simplest sum of products for F(A, B, C) = Σm(0, 2, 3, 7).

34. A function of A, B, C is 0 only in the rows of maxterms M(0, 1, 4, 5) and 1 in every other row. Write the simplest expression for F.

35. Add the binary numbers 1100111 + 101110. Give the sum in binary and in decimal.

36. A car buzzer F sounds when the key is in the ignition and either the door is open or the seatbelt is not buckled. Let A = 1 when the key is in, B = 1 when the door is open, C = 1 when the seatbelt is buckled. Write F as a sum of products.

37. In a hands-on task, a known 2.20 kΩ resistor and a mystery resistor are in series with a battery. You may only measure voltage. The known resistor has 4.62 V across it and the mystery resistor has 2.88 V. Find the mystery resistance.

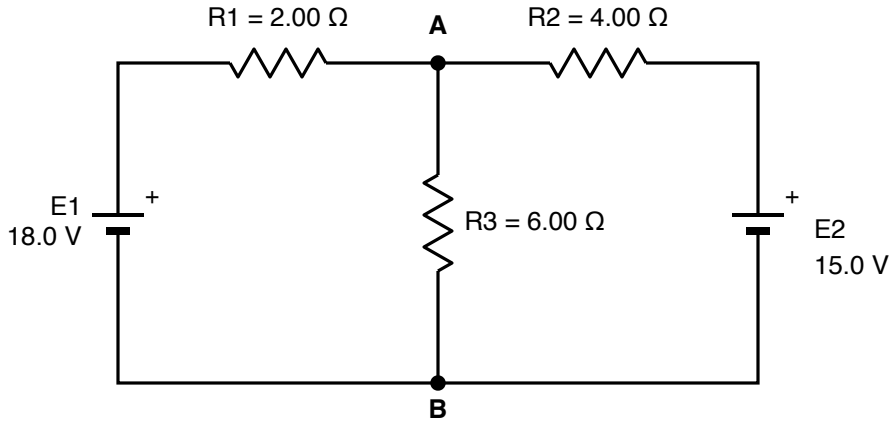
38. A USB charger label lists 5 V ⎓ 3 A, 9 V ⎓ 2 A and 12 V ⎓ 1.5 A. (a) Give the power of each level. (b) A 4500 mAh, 3.87 V phone battery charges at 18 W with 85% efficiency. About how long does a full charge from empty take?

Section III: Circuit problems

Show your work. 2 points per part.

Problem A: two batteries (Kirchhoff's laws) (Questions 39–42)

E1 = 18.0 V and E2 = 15.0 V are ideal batteries (the long plate, +, is at the top of each). R1 = 2.00 Ω, R2 = 4.00 Ω, R3 = 6.00 Ω. Let I_1 be the current in R1 flowing toward node A and I_2 the current in R2 flowing toward node A.



39. Use KCL at A, then KVL around the left loop and the right loop. Write the two equations in I_1 and I_2.

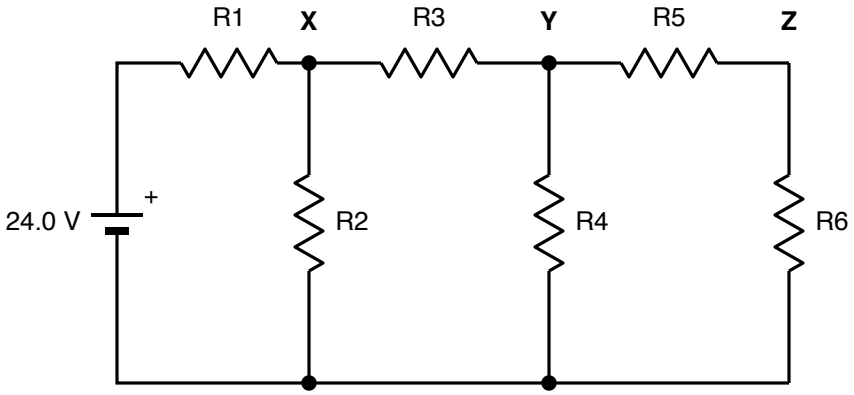
40. Find I_1.

41. Find I_2. Is E2 giving power to the circuit or being charged?

42. Find the voltage of A relative to B and the power used by R3.

Problem B: ladder network (Questions 43–45)

A 24.0 V ideal battery feeds the ladder below. The series resistors R1, R3 and R5 are each 1.00 kΩ. R2 and R4 are each 2.00 kΩ. R6 is 1.00 kΩ. X, Y and Z are the top nodes.



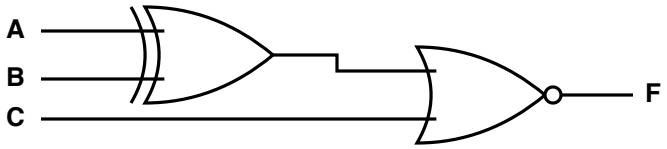
43. Find the equivalent resistance that the battery sees.

44. Find the current in R2.

45. Find the voltage across R6 (node Z to the bottom wire).

Problem C: logic circuit (Questions 46–47)

An XOR gate and a NOR gate are connected as shown. The inputs are A, B, C; the output is F.

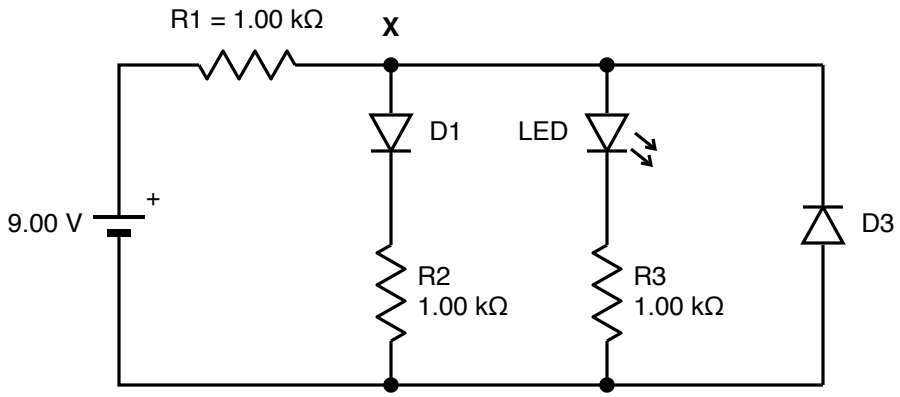


46. Write F as a Boolean expression, then as a sum of products.

47. List every input row ABC that makes F = 1 (and give the minterm numbers).

Problem D: diodes in parallel branches (Questions 48–50)

A 9.00 V ideal battery feeds node X through R1 = 1.00 kΩ. From X to the bottom wire there are three branches: a silicon diode D1 (0.70 V when on) with R2 = 1.00 kΩ; a red LED (2.00 V when on) with R3 = 1.00 kΩ; and a silicon diode D3 with its anode on the bottom wire.



48. Which of D1, the LED and D3 conduct? Explain in one line.

49. Find the voltage of node X.

50. Find the current in the LED.

End of the test. Check your answer sheet.

Answer sheet · Practice Test 4 (State level)

Team name: _____ Team number: _____

Multiple choice: circle one letter. Other questions: write the answer with units.

1. A B C D
2. A B C D
3. A B C D
4. A B C D
5. A B C D
6. A B C D
7. A B C D
8. A B C D
9. A B C D
10. A B C D
11. A B C D
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